

ASSUMPTIONS.md — implementation-level decisions requiring owner review

Companion to `hexis_ctree` v0.1.0. Session date: 2026-07-07. Reference texts: HEXIS Master Spec v2.0 (cited as §N), Decision Log v2.0 (D01–D39), and Schürmann & Grassberger 1996 (arXiv:cond-mat/0203436, full text fetched and read in this session; cited as S&G with equation/section numbers).

Legend — each item carries one status:

- **[IMPLEMENTS FROZEN]**: direct implementation of an already-frozen decision; listed for traceability, no ratification needed.
 - **[READING — RATIFY]**: the spec/S&G leave an operational detail open or ambiguous; this codebase adopts one reading. Requires an explicit Decision-Log entry (or a note under an existing D) before G2.
 - **[T5.1 MATERIAL]**: a documented divergence between the frozen C0 defaults (D18) and the S&G-published choices — exactly what Task T5.1 must collect for the possible D18-A1 amendment and for the SG96-correspondence note in the Methods section.
 - **[PENDING]**: spec-mandated behaviour not yet implemented in this package; tracked so it cannot silently disappear.
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A. Frozen decisions implemented

A1. Single instrument (D32). One model class; all quantities produced here (`h_online`, `CE`, `gain`, `depth`) are readings of fitted context trees. No retired estimator (per-chunk entropy, standalone conditional entropy, MI, chunk JSD, PERMANOVA, LZ) exists anywhere in the package. **[IMPLEMENTS FROZEN]**

A2. C0 defaults (D18/D37, §4.1/§6.3). `ContextTreeConfig()`
`== ContextTreeConfig.hexis_c0()` ($\beta = 0.5$, $k_{\min} = 2$, $\gamma = 0.0$ bits, $d_{\max} = 8$, growth on first visit, monotone-stop selection). The Latin $d_{\max} = 6$ (D24) is exposed as `hexis_c0(max_depth=6)`. Tested. **[IMPLEMENTS FROZEN]**

A3. Sentence-level hard reset (D10, P-RESET primary). Contexts truncate at sentence start; no cross-sentence statistics; no BOS symbol in the alphabet; averages over real tokens only; `available_past(i) = i`. The P-BOUND sensitivity arm (boundary symbol #, D10/D37) is **not** in this package — it belongs to the alphabet/pipeline layer, where # must be added to the frozen alphabet before fitting. **[IMPLEMENTS FROZEN]** (P-BOUND: **[PENDING]** at the pipeline layer.)

A4. Position restriction (D35). `MIN_AVAILABLE_PAST_FOR_GAIN_AND_DEPTH = 4`; `gain` and `depth` aggregators default to it; the P1 ingredient (`mean_codelength_bits`, `cross_entropy`) uses all real positions, per §4.4 ("ΔCE needs no restriction"). The unrestricted variant is reachable by passing `min_available_past=0` (one D37 sensitivity cell). **[IMPLEMENTS FROZEN]**

A5. Label-free P2/S1 construction (D36). `pooled_document_scores` takes a fitted pooled model and a mapping `doc_id → document`; no regime information exists in its signature or body. Per-document statistics are computed once and permutations regroup fixed values. `test_scores_are_label_free_and_reproducible` asserts identity of repeated scoring; note that the spec's stronger byte-identical registry-permutation test (§7, `test_scores.py`) applies to the future `pooled_scores` orchestration function that reads a registry — **[PENDING]** at that layer. **[IMPLEMENTS FROZEN]** for the engine-level guarantee.

A6. Held-out primacy (D19/D32). `h_online` is exposed as a descriptive property with the Eq. (28) caveat in its docstring; the frozen `cross_entropy / score_document` path is the inferential surface. **[IMPLEMENTS FROZEN]**

A7. Units and bases (§0.3). $\log_2$ everywhere; all quantities in bits; tested against hand-computed values. **[IMPLEMENTS FROZEN]**

A8. Determinism (D26-compatible). Fitting has no randomness; the alphabet ordering is a pure function of the symbol set; every synthetic generator and the Monte Carlo permutation take explicit seeds. The manifest/seed-derivation machinery of §6.4–§6.5 (global seed 20260706 XOR `crc32(analysis_id)`, manifest.json sidecars) is repository infrastructure, not engine logic: **[PENDING]** at the pipeline layer.

B. Readings of the spec adopted here — ratify before G2

B1. Selection descent = "monotone-stop" operationalized as monotone descent. §4.1: "`s` = deepest `sj` with, for every $i = 1 \dots j$, $\text{total}(s_i) \geq k_{\min}$ and $\Delta(s_i) > \gamma$ (monotone-stop)". Implemented as: walk from the root; step to the matching child iff it satisfies both conditions; stop at the first failure. This selects the deepest prefix whose *entire chain* qualifies — the literal quantifier in the spec sentence. S&G's own wording (Sec. V.B: shortest context with $\Delta < -\delta$ whose child does not improve) coincides with this in unambiguous cases; the gap case $\Delta_{\text{child}} \in (-\delta, 0]$ is resolved conservatively (stop at parent). Propose recording this one-paragraph reading under D18 (no parameter changes). **[READING — RATIFY]**

B2. `k_min` applied identically online and frozen. §4.1 defines `k_min` in the selection rule; §4.5 says `h_online` uses "online Δ -based selection". Adopted reading: the online selection applies the full rule (Δ and `k_min`) on pre-update totals; frozen evaluation applies it on final training totals. A node with `total = 1` (just created) is therefore never selected online under C0 — coherent with the "earn its keep" logic of §4.1. **[READING — RATIFY]**

B3. Growth = "eager", at most ONE new node per position — a declared deviation from the literal §4.1 pseudocode. The spec's training pass (`node.children.setdefault( $\sigma$ , new Node)` inside the depth loop) grows the *whole* matched-extension chain, up to `d_max` new nodes per position. Implemented instead: extend the matched path by exactly one node per position (the S&G/Rissanen construction, kept under both policies), with "eager" = create on first visit, "rissanen" = create when the deepest-matched count of the current symbol reaches 2. **These are NOT selection-equivalent under `k_min = 2`:** for a deep context occurring only twice in the corpus, the literal chain growth reaches `total = 2` at full depth on the second occurrence (selectable, given $\Delta < -\gamma$), whereas the one-node-per-position policy needs roughly (depth) occurrences to build the

chain that far. Direction of the effect: the implemented policy is strictly more conservative for *rare deep contexts* (fewer weakly-supported deep selections; frequent contexts converge to the same tree), and it bounds the node budget by N instead of $N \cdot d_{\max}$. Trade-off to decide at ratification: fidelity to the spec pseudocode (literal chain growth, a ~ 5 -line change) versus fidelity to S&G's frugal construction plus conservatism (current). Flag for the mathematical-supervisor touchpoint; candidate D18-A1 item either way. **[READING — RATIFY]**

B4. Δ sign convention. The spec stores $\Delta(s) = L_{\text{par}} - L_{\text{self}}$ ("bits saved", positive good); the code stores the S&G-oriented $\text{delta} = L_{\text{self}} - L_{\text{par}}$ (negative good). $\Delta_{\text{spec}} = -\text{delta}$; the acceptance condition $\text{delta} < -\gamma$ is exactly $\Delta_{\text{spec}} > \gamma$. Tested equivalence; the context-lexicon output (D38, future) must report $-\text{delta}$ as $\Delta(s)$. Documentation-level item; propose a one-line note under D18. **[READING — RATIFY]**

B5. Out-of-alphabet symbols raise; in-alphabet unseen symbols are β -smoothed. §4.3 requires unseen symbols to get positive probability (satisfied by Eq. (21) for every symbol of the frozen alphabet) and the project requires fail-loud behaviour on malformed input (§0.6, handoff rules). Adopted split: a symbol absent from the frozen alphabet is a data error (raise with the offending symbol); a symbol in the alphabet but never observed in a context is smoothed. Consistent with alphabet freezing at G1. **[READING — RATIFY]** (likely uncontroversial).

B6. P2/S1 aggregation = unweighted mean over documents. §4.4: " $P2 = \text{mean}_{\{d \in \text{HEX}\}} G(d) - \text{mean}_{\{d \in \text{PROSE}\}} G(d)$ " — plain document means, matching the document-level randomization unit (D21). `document_mean_difference` is the default statistic of the permutation test; the token-weighted variant is retained, explicitly labeled non-spec/descriptive.

[IMPLEMENTS FROZEN] (listed here because the first draft of this layer used the token-weighted form; corrected in this session.)

B7. Transfer matrix is a descriptive display. `transfer_matrix` pools LODO folds token-weighted for a compact CE(row|column) table (an F2-style display). The confirmatory P1 of §4.4 is *not* this matrix: it is the unweighted mean over evaluation documents of $\Delta \text{CE}(d) = \text{CE}(d|\text{other}) - \text{CE}(d|\text{own})$ under protocol (b) with T^* -matched training pools and $S = 20$ seeds — which requires the subsampling machinery of the protocols phase. **[READING — RATIFY]** for the display; confirmatory P1: **[PENDING]**.

C. C0 ↔ S&G-published divergence map (Task T5.1 feeder)

Collected while reading S&G §IV–§V against D18. On the three synthetic validation sources the two presets give near-identical results (README table) — the divergences below are design choices, not observed distortions.

#	Parameter	C0 (frozen, D18)	S&G published	Note
C.1	β	0.5 (KT-style)	$1/d$, Eq. (30)	S&G's $1/d$ is an <i>empirical</i> finding on English text, not a theorem; β is a D37 sensitivity axis ($\{1/ A , 0.25,$

#	Parameter	C0 (frozen, D18)	S&G published	Note
				1.0}))
C.2	Count floor k_min	2	none ($\equiv 1$)	HEXIS addition; S&G's growth rule (a node exists only after a repeat) partially overlaps its effect but is a different mechanism
C.3	Growth policy	on first visit ("eager")	on second occurrence ("rissanen", Sec. IV end)	both implemented; one-node-per-position in both (see B3)
C.4	Depth cap	d_max = 8 (Latin 6)	none in §V; truncated trees sanctioned in §VI.B as upper bounds	cap is also a D37 axis ({6, 12})
C.5	Threshold	$\gamma = 0$ bits	$\delta \geq 0$, "chosen differently in different applications" (§V.B)	compatible; $\gamma = 0$ is the permissive end
C.6	Boundary handling	sentence hard reset (D10)	none (continuous sequences)	HEXIS addition; no S&G counterpart
C.7	Evaluation mode	frozen out-of-sample scoring	online/adaptive only	HEXIS addition required by P1/P2/S1 (D19/D36); S&G's $\hat{h}$ retained as h_{online}

Suggested use: C.1–C.5 become the body of the SG96-correspondence note in Methods (§4.1 already announces it) and, if the owner wishes, a single D18-A1 amendment recording that the published-S&G preset is available as `ContextTreeConfig.sg96_published()` and behaves indistinguishably on the analytic validation suite.

D. Pending (spec-mandated, deliberately out of this package)

- **D-a.** Protocols (a)/(b)/(c) orchestration: T^* computation, whole- sentence subsampling to T^* , $S = 20$ seeds with the §6.5 seed derivation, learning curves (F8). → `protocols/` module, Phase 4 of the roadmap.
- **D-b.** Exact inference: enumeration of the 2^{11} sign-flips and the $C(11,5) = 462$ label

assignments (§5.1); the Monte Carlo shuffle here is an engine-validation stand-in and must not be used confirmatorily.

- **D-c.** `select = argmax` variant (one D37 cell).
- **D-d.** P-BOUND boundary arm (alphabet-level #, D10/D37).
- **D-e.** Spec §7 test names/fixtures (four analytic processes **as specified there**: uniform 2.000; period-3 CE < 0.02; Markov `p_stay` = 0.8 → 0.7219; order-2 XOR ϵ = 0.1 → 0.4690 vs $\approx$ 1.0). The present suite covers uniform (2.000 ✓), an order-1 chain (0.4690 via `p_stay` = 0.9 ✓) and a depth-2 template; the period-3 and XOR fixtures at the spec's exact parameters should be added verbatim when the repository skeleton (§6.1) is created, so that gate G3 is checked against the spec's own numbers.
- **D-f.** Slice-identity diagnostics (§4.6): root = add- β unigram (< 1e-12), depth-1 nodes = smoothed bigrams, evaluate-on-train consistency. The quantities exist in the model; the dedicated `diagnostics.py` with the three machine checks is repository work.
- **D-g.** Context lexicon (D38): rank nodes by total importance $\Delta_{\text{spec}}(s) = -\text{delta}$; `node_table / top_contexts` interfaces per §6.2.
- **D-h.** Manifests and seed derivation (§6.4–§6.5).

None of the pending items blocks the validity of the engine; all are tracked so that the repository phase starts from an explicit list.